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Predictive analytics in autonomous vehicles safety: crash outcome modeling

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ABSTRACT

Traffic-related fatalities, often caused by human error, present a global challenge. Autonomous vehicles (AVs), designed to reduce these fatalities by eliminating human error, still experience crashes. Prior research, limited by the scope of the data set and analytical depth, has not fully explored these factors, resulting in a lingering lack of clarity about the cause of the crashes. The study utilized a comprehensive dataset spanning from 2014 to May 2024 and implemented a stratified sampling alongside class weighting techniques to rectify imbalances within the crash data. A variety of machine learning algorithms such as logistic regression, decision trees, random forests, gradient boosting, bagging classifiers, and easy ensemble methods were utilized to perform the data analysis. The random forest model stood out for its balance in sensitivity and specificity, reducing false negatives. The study revealed that factors such as the extent of vehicle damage, the AV manufacturer, and type of collision were critical predictors of crash injuries. The research provides valuable insights for various stakeholders, including transportation professionals, AV manufacturers, and policy-makers. The findings can guide the development of enhanced safety protocols, improved vehicle designs, and more informed policy frameworks aimed at reducing AV-related injuries.

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Autonomous vehicles; crash injury; predictive modeling; crash patterns; machine learning; road accidents

1. Introduction

Traffic fatalities are one of the leading causes of death globally (Ashraf et al., 2021). An estimated 90% of them are caused by human error, with 40% a direct result of driving under the influence or while distracted or fatigued (Durić & Miladinov-Mikov, 2008; Schanz & Hamann, 2012; Sing, 2018). Autonomous vehicles (AVs) have the potential to significantly reduce crash frequency and injuries through the elimination of human error within the driving environment. In addition to improved roadway safety, widespread deployment of AVs can reduce traffic congestion, minimize energy consumption and

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emissions, improve air quality, and significantly impact land use (Almaskati, Kermanshachi, et al., 2024; Houseal et al., 2022).

Autonomous driving technologies are designed to assist human drivers by assuming the control of driving tasks (Khattak et al., 2020), which varies by type of vehicle. The Society of Automotive Engineers (SAE) has defined six levels of automation: Level 0 has zero automation, Levels 1 to 3 have partial automation, and Levels 4 to 5 have full automation (Favarò et al., 2017). Although AV technology promises to resolve the issue of human driving error, it is still in the developmental stage and may pose other challenges. Consequently, to ensure their safe operation, regulatory requirements for testing AVs have been introduced, and multiple states are complying by conducting extensive testing (Patel et al., 2023; Ren et al., 2022; Sinha et al., 2021). Manufacturers in California are permitted to test AVs on public roadways on the condition that reports on disengagements and collisions involving them are submitted to the California Department of Motor Vehicles (CA DMV), who subsequently compile and publish the data (Channamallu et al., 2023; Liu et al., 2021). Since these reports are the only available datasets on AV performance, researchers from all around the world have evaluated the information, extensively examining the performance of AVs through the use of a variety of statistical methods.

To identify the collision patterns and the factors contributing to crash severity, researchers have employed various methods including binary logistic regression (Xu et al., 2019), ordinal regression (Wang et al., 2019), descriptive analysis and Fisher's exact test (Ye et al., 2021), the Bayesian latent class model (Das et al., 2020), descriptive statistics (Leilabadi & Schmidt, 2019; Pamidimukkala et al., 2023), Bayesian networks (Kutela, Avelar, et al., 2022), and the Bayesian-ordered probit model (Yuan et al., 2022). These models were designed to examine the data and investigate AV crash severity factors; however, such statistical models may be characterized by class imbalanced datasets. This means that the data set contains a significantly higher number of non-injury crashes compared to injury crashes, leading to an uneven distribution of crash classes. Such imbalances can cause statistical models to be biased towards the majority class, making it difficult to accurately predict the minority class (injury crashes). Furthermore, while the studies' findings illuminate the behavior of AVs, they are constrained by limited data sets and fail to encompass all currently available CA DMV data (Channamallu et al., 2024b; Yuan et al., 2022).

In recent years, the AV industry has experienced rapid technological advancements, from the development of more advanced sensors to improved AI-driven decision-making systems. As AV technology evolves, it becomes increasingly important for research to incorporate the most current data and insights. Despite previous research, there remain significant gaps in understanding the factors that influence crash outcomes involving AVs, particularly due to the use of traditional statistical methods and limitations in data scope, which may not fully reflect the latest technological improvements. Moreover, the challenge of class imbalance in crash data has not been adequately addressed. To fill these gaps, this study aims to not only predict crash outcomes but also identify and rank the contributing factors by severity using advanced machine learning techniques and the most comprehensive dataset available from the California DMV. The primary objectives of this study are to: (1) develop predictive models by using machine learning algorithms that can effectively differentiate between injury and non-

injury crashes involving AVs, and (2) **rank the contributing factors**, based on their predictive importance, to understand their role in crash outcomes. The findings will be beneficial to transportation professionals, AV manufacturers, and policymakers by providing insights into the performance of AVs on public roadways. Additionally, the results may be utilized as a reference for designing strategies and policies that can mitigate the factors contributing to crash severity.

2. Literature review

Previous studies have extensively explored AVs' performance on public roadways to provide insight on how to best ensure their safe operation. Since 2014, California has mandated that all AV manufacturers release AV disengagement and collision reports and has made that information available to the public through the CA DMV (Guo & Zhang, 2022). Some other countries permit AV testing, but their data is not yet widely accessible (Almaskati, Pamidimukkala, et al., 2024; Houseal et al., 2022). The open-source availability of the CA DMV data has allowed researchers to study collisions through the lens of specific objectives by employing a variety of statistical models and descriptive analyses to scrutinize AV performance and explore safety concerns. Specifically, the data has been analyzed to identify crash severity factors (Zhu & Meng, 2022), compare the performance of AVs with that of conventional vehicles (Petrović et al., 2020), determine the patterns of crashes (Song et al., 2021; Xu et al., 2019), evaluate the maturity of AV driving systems (Guo & Zhang, 2022), and determine the root cause of AV-involved accidents (Channamallu et al., 2024a; Leilabadi & Schmidt, 2019).

A significant number of recent studies have attempted to identify the risk factors associated with collisions involving AVs. A study by Xu et al. (2019) investigated the characteristics and patterns of connected and AV-involved crashes through an ordinal logistic regression method. They considered the crash location, driving mode, weather conditions, vehicle movements prior to the crash, collision type, speed, crash severity, and location of the damage and determined that rear-end collisions, one-way roads, and roadside parking were the primary factors contributing to the severity level of crashes. Similarly, Das et al. (2020) evaluated collision reports to study the frequency of collisions and the related contributing factors, using a Bayesian latent class model to identify classes of collision patterns and variables such as collision type, weather conditions, lighting conditions, vehicle damage, and whether the vehicle was stationary or moving. The results revealed that a high proportion of **severe injuries were associated with turning movements**, multi-vehicle collisions, poor lighting conditions, and sideswipe and rear-end accidents. Ye et al. (2021) supported these findings and through examination of the patterns of road traffic injuries involving AVs, **identified poor lighting conditions** as the most significant factor. Although these studies provide useful insights into the performance of AVs and their crash severity factors, the literature lacks a comparative analysis of models on the CA DMV crash data. In their study, Xu et al. (2019) utilized bootstrap based logistic regression to investigate collision factors, whereas Ye et al. (2021) and Das et al. (2020) applied a binary logistic regression and Bayesian latent class model, respectively, for their evaluation purposes. The findings of these studies can be enhanced through the consideration of multiple models to ensure that the most predictive model is developed. Moreover, statistical methods alone may not effectively capture the nuances

between the contributing factors that were evaluated, thus the analysis may be strengthened through the inclusion of machine learning models.

Wang et al. (2019) investigated the contributing factors and mechanisms of AV crashes, utilizing the crash data base and statistical modeling approaches such as ordinal logistic regression and CART classification tree, and found that crashes are considerably more severe when the AV is primarily responsible. In contrast, when the AV is not the faulty party, the severity level was found to be significantly lower. They also determined that individuals are more likely to suffer severe injuries when accidents occur on highways, possibly due to the high travel speeds. While this study's methodology incorporated both statistical methods and machine learning, the findings are constrained by the data, in which only 113 crash records were included dating from 2014 to 2018 (Wang et al., 2019). One study considered the ability of AVs to recognize different situations by analyzing how their performance was impacted by environmental factors. The Pearson chi-square test was conducted and revealed that roadway surfaces have the most significant level of association with AV performance (Leilabadi & Schmidt, 2019); e.g. AVs are more likely to be involved in a collision when the road is wet. Contrastingly, Zhang and Xu (2021) classified down-slopes, nighttime driving, multiple-vehicle accidents, and high-density traffic as the primary contributing factors to an increased likelihood of a severe crash. Similarly, the findings may be hindered by data limitations and may not be reflective of the current performance of autonomous driving systems due to the rapid evolution of AV technology.

Such limitations are addressed in more recent publications, including a study by Ding et al. (2023) which utilized a multinomial logit model to evaluate the variables that impact crash severity and learned that the most contributing factors are weather conditions, traffic incidents, and type of collision. However, the model estimation results are limited by the implementation of one modeling approach and may be enhanced through comparative analysis. A study by Kutela, Das, et al. (2022) conducted a comparative analysis of various machine learning algorithms to explore the impact of AV crashes on vulnerable road users. Although the study provides useful insights, its application of text mining techniques is focused on the evaluation of the involvement of vulnerable road users in AV collisions rather than AV crash severity factors. In an analysis of crash reports using Bayesian networks, Kutela, Avelar, et al. (2022) considered vehicle at fault, collision type, and injury outcomes to determine how each of those variables impacted AV crashes. While the findings highlight the predominance of rear-ended collisions and a higher likelihood of injuries when the AV was at fault, the study's findings could be strengthened through the consideration of additional variables that might impact AV collisions.

In summary, previous studies have extensively analyzed AV crash data, focusing on factors influencing crash outcomes (Das et al., 2020; Xu et al., 2019). However, many of these studies rely on older datasets, which do not account for recent technological advancements in AV systems (Leilabadi & Schmidt, 2019). For instance, improvements in sensor technology, machine learning algorithms, and AV software over the past few years are not reflected in the earlier datasets used by these studies. This limitation creates a gap in the research, as current crash outcomes may be influenced by the latest innovations in AV technology. Additionally, the challenge of class imbalance – where

the majority of records are non-injury crashes – has not been adequately addressed in many studies (Wang et al., 2019; Xu et al., 2019), potentially leading to biased results.

3. Methodology

This section describes the comprehensive process undertaken to analyze and predict outcomes of AV crashes. It outlines the systematic approach from initial data collection to the selection of the best performing model and determination of factors that most affect the outcome of a crash. The methodology adopted for this study is presented below in Figure 1.

Data for the study was gathered from reports submitted by testing companies to the CA DMV and encompassed various attributes of specific vehicles, as well as environmental, traffic and other contextual factors. The data was pre-processed, including cleaning, normalization, and transformation of data, to ensure quality and consistency, and outliers were addressed, and missing values were removed to maintain the integrity of the dataset for model development. A thorough selection process was implemented to identify the most relevant variables for the model and resulted in a refined set of predictive features. Multiple predictive models were developed, and each model was tuned to optimize its parameters for the best

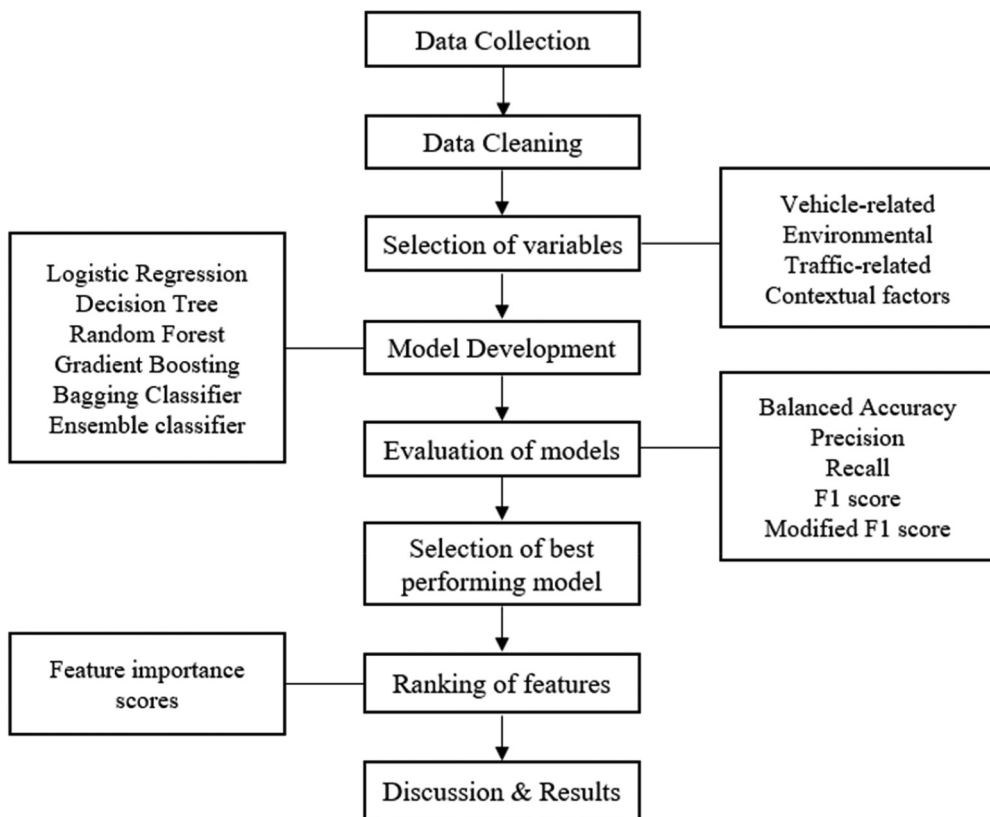


Figure 1. Methodology.

predictive performance and was evaluated using a range of metrics, such as accuracy, precision, recall, and the F1 score. The model that exhibited the highest performance across various evaluation metrics was selected as the best performing model and its predictive capability was deemed superior for predicting the possibility of injuries resulting from AV crashes. The feature importance score was analyzed to rank the variables based on their influence on the model’s predictions and highlighted the most critical factors in predicting crash outcomes. A discussion was conducted to interpret the significance of the top-ranked features, and the findings from the best performing model were compiled, presenting a clear and evidence-based hierarchy of features that contribute to the likelihood of injuries from AV crashes.

3.1. Data set

The study’s data was sourced from the CA DMV, which requires that operators conducting on-road testing of AVs report crashes in PDF form within 10 business days. This analysis involved collecting details from each PDF report on the CA DMV’s website from 2014 to May 2024, extracting the data, and organizing it into an Excel spreadsheet for quantitative analysis. [Figure 2](#) presents the yearly distribution of AV crashes in the data set.

In the early years, from 2015 to 2017, crash percentages remained relatively low, likely reflecting the limited deployment and initial testing phases of AVs. By 2018 and 2019, there was a noticeable increase in AV crashes, corresponding to the growing presence of AVs on public roads and expanded testing. A decline in 2020 can likely be attributed to reduced traffic during the COVID-19 pandemic. In the subsequent years, from 2021 to 2023, crash percentages stabilized at a higher level, suggesting that AV activity and reporting became more consistent as these vehicles became more prevalent. The drop observed in 2024 is most likely due to incomplete data for the year. Overall, the data underscores the growing

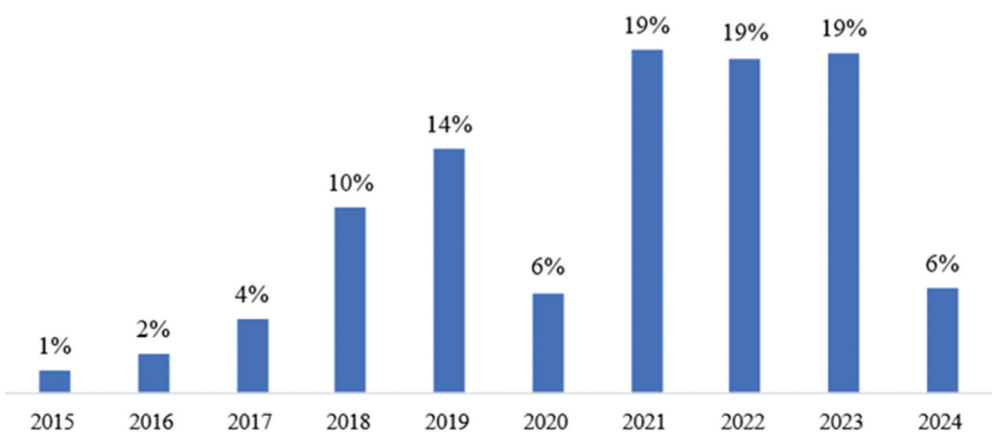


Figure 2. Yearly distribution of AV crashes.

presence of AVs on the roads, accompanied by a corresponding rise in reported crashes, especially during periods of increased testing and deployment.

3.2. Data cleaning

To ensure the quality and integrity of the dataset, several data cleaning steps were performed. First, crashes with numerous missing values, particularly in key variables such as crash severity, vehicle type, and environmental factors, were excluded from the dataset. This step was crucial for maintaining the reliability of the data and preventing missing values from skewing the results of the analysis. Next, statistical methods, including Z-scores, were applied to detect outliers in the dataset. These outliers were reviewed, and extreme values deemed erroneous or unrepresentative of the general population were removed. This helped avoid distortion in the analysis and ensured that the results reflected typical crash scenarios. Finally, certain categorical variables, such as weather conditions, vehicle types and road types, were recoded into binary or ordinal variables. This transformation was done to simplify the analysis while preserving meaningful distinctions between categories. These steps ensured that the dataset was clean, consistent, and ready for further analysis.

After cleaning the data, the data set included 530 individual incident records and 16 variables. The variables pertain to the characteristics of the AV, the other party involved in the incident, prevailing traffic conditions, and other specifics noted in the crash report. Descriptive statistics for these variables are presented in Table 1.

3.3. Model development

The crash model developed for this study incorporated 16 independent variables and a dependent variable that denotes the prediction of injury resulting from a crash. The binary dependent variable was categorized into 0 for non-injury crashes, and 1 for injury-related crashes. A significant disparity in sample size was observed between the two classes, with 483 instances in the non-injury category and only 47 in the injury category, resulting in an imbalanced data distribution. To address the issue of imbalance, the models incorporated class weighting strategies, assigning inversely proportional weights to each class based on the frequency with which they were entered in the dataset. This method is a cornerstone for mitigating issues of imbalance, as it heightens the influence of less represented classes to foster a more equitable learning process. Class weight uses the formulae presented in Equation 1, where n_{sample} is the number of samples, n_{classes} is the number of classes in the dataset, and $\text{np.bincount}(y)$ is the total count of a specific class in that dataset.

$$\text{Class weights of a specific class } y = \frac{n_{\text{sample}}}{n_{\text{classes}} \times \text{np.bincount}(y)} \quad (1)$$

The data was then split into training and testing sets, with 80% allocated for training and 20% for testing. To address the skewness resulting from an uneven class distribution, the study utilized stratified sampling when dividing the dataset, which preserved the proportion of each class across training and testing sets. Several models, both with and without class weights, were developed and tested to discern the most effective method for handling the skewed data distribution. Table 2 provides a comprehensive summary of the

Table 1. Descriptive statistics of variables.

Variable	Type	Definition	Descriptive statistics
Vehicle type	Categorical	=1 Two wheels (i.e. bicycle, scooter) =2 Sub-compact/compact cars =3 Mid-size cars =4 Trucks/buses/trailers	F(1) = 9% F(2) = 34% F(3) = 52% F(4) = 5%
Road type	Categorical	=1 Street =2 Avenue =3 highway =4 Boulevard =5 Freeway	F(1) = 76% F(2) = 19% F(3) = 1% F(4) = 2% F(5) = 2%
Intersection Geometry	Categorical	=0 Straight Road =1 Y-intersection =2 T-intersection =3 Cross intersection =4 Complex intersection	F(0) = 7% F(1) = 9% F(2) = 12% F(3) = 65% F(4) = 7%
Signal	Binary	=1 Signalized =0 Non-signalized	F(1) = 53% F(0) = 47%
Parking provision	Binary	=1 Parking provision =0 No Parking provision	F(1) = 62% F(0) = 38%
Vehicles involved	Categorical	Number of vehicles involved	F(1) = 1% F(2) = 97% F(3) = 2%
Mode	Binary	=1 Autonomous =0 Manual	F(1) = 57% F(0) = 43%
AV Status	Binary	=1 moving =0 stopped	F(1) = 51% F(0) = 49%
Other party's vehicle status	Binary	=1 moving =0 stopped	F(1) = 93% F(0) = 7%
Damage to AV	Categorical	=1 None =2 minor =3 Moderate =4 Major	F(1) = 9% F(2) = 70% F(3) = 19% F(4) = 2%
Weather	Binary	=1 Clear =0 Less visibility	F(1) = 88% F(0) = 12%
Collision type	Categorical	=1 head-on =2 Swipe =3 Rear-end =4 hit object =5 Freeway	F(1) = 18% F(2) = 39% F(3) = 40% F(4) = 3% F(5) = 0%
Lighting	Binary	=1 Day lighting =0 Poor lighting	F(1) = 73% F(0) = 27%
Road Surface	Binary	=1 Dry =0 Wet	F(1) = 95% F(0) = 5%
Movement preceding collision	Categorical	=1 Stopped =2 Proceeding straight =3 Others	F(1) = 48% F(2) = 26% F(3) = 26%
Company	Categorical	Make of the AV vehicle	Multiple companies

models employed in this study along with basic setup parameters used in modeling. These models were chosen for their ability to handle the complex and often imbalanced data typically found in AV crash analysis.

3.4. Evaluation metrics

In evaluating the model, traditional accuracy metrics were set aside for more nuanced measures like balanced accuracy, as they offer a truer reflection of model performance in skewed datasets. Accuracy indicates the proportion of correctly classified instances out of

Table 2. Models utilized for the study.

Model	Description	Application in AV crash prediction	Basic setup parameters
Logistic Regression	A statistical model used for binary classification, it estimates the probability of a binary response based on one or more predictor variables.	Utilized for predicting binary outcomes like crash occurrence or injury severity in AVs	Penalty: l2 C: 1.0 (default regularization strength) Random state: 42
Decision Tree	A model that uses a tree-like graph of decisions and their possible consequences and is used for classification and regression	Helps identify critical decision nodes and variables that influence AV crash outcomes	Criterion: gini Splitter: best Max depth: None Min samples split: 2 Random state: 42
Gradient Boosting	An ensemble technique that builds the model in a stage-wise fashion, often used for classification and regression problems	Effective for improving prediction accuracy in AV crash data by sequentially correcting errors of previous models	Loss: deviance Learning rate: 0.1 Number of estimators: 100 Max depth: 3 Random state: 42
Random Forest	An ensemble learning method that operates by constructing multiple decision trees at training time	Used for its robustness and accuracy in predicting various outcomes related to AV crashes	Number of estimators: 100 Criterion: gini Max depth: None Min samples split: 2 Random state: 42
Balanced Random Forest	A variation of the Random Forest model that adjusts for imbalanced data sets	Specifically addresses the class imbalance issue common in AV crash data	Base estimator: Decision tree classifier Number of base estimators in the ensemble: 10 Random state: 42
Bagging Classifier	An ensemble of meta-estimator fitting base classifiers on random subsets of the original data set	Useful in reducing variance and avoiding overfitting in AV crash prediction models	
Balanced Bagging Classifier	A modified Bagging Classifier that addresses class imbalance by altering dataset distribution	Tailored for AV datasets where certain crash outcomes are rare but significant	
Easy Ensemble Classifier	An ensemble of classifiers, each trained on a different balanced bootstrap sample	Enhances predictive performance in scenarios with highly imbalanced AV crash data	

the total instances, as shown in Equation 2, where TP represents true positives, TN represents true negatives, FP represents false positives, and FN represents false negatives.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Precision, as defined in Equation 3, measures the ratio of true positive predictions to the total number of positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (3)$$

Recall, illustrated in Equation 4, indicates the proportion of true positive observations relative to the total actual positive instances in the dataset

$$Recall = \frac{TP}{TP + FN} \quad (4)$$

The model's ability to reduce false positives was prioritized due to the critical nature of falsely predicting non-injury incidents; hence, a **modified evaluation method suggested in previous studies was adopted, with the modified F1 score serving as the key metric** (Li et al., 2018; Sinha et al., 2021). This score strikes a balance between precision and recall,

ensuring robust predictions for both classes, which is vital when dealing with imbalanced data. The formula for the modified F1 score is presented in equation 2.

$$\text{Modified F1 score} = (1 + \beta^2) \times \frac{\text{precision} \times \text{recall}}{\beta^2 \times (\text{precision} + \text{recall})} \quad (5)$$

where β is a parameter; when β is a parameter with a value between 0 and 1, the focus is more on precision, whereas a β greater than 1 indicates that recall is more important. In this study, a β of 0.5 was employed to place greater importance on precision, due to the use of false positive, which is a significant factor in evaluating the performance of the model.

4. Results & discussion

The evaluation of various crash prediction machine learning models revealed a broad spectrum of performance. Table 3 shows the confusion matrix, and a suite of evaluation metrics (balanced accuracy, precision, recall, the conventional F1 score, and the modified F1 score) for each model. This array of statistics offers a thorough insight into the predictive power and accuracy of each model. These results illustrate the complexity of predictive modeling in imbalanced datasets, particularly in the context of predicting traffic crashes. Class weights and specialized ensemble methods can improve balanced accuracy and recall, but they often compromise precision. The models' tendency to have a few FNs is concerning in real-world scenarios, as it could mean overlooking potential injury-causing incidents.

Logistic Regression without class weights was rejected despite its high recall because it exhibited a lack of true negatives and lower precision. This imbalance indicates that the model struggled with FPs, which is critical in predictive modeling of crashes where precision is vital. Logistic Regression with class weights improved precision but significantly compromised recall, leading to a higher number of FNs. In safety-related predictions, missing actual incidents is unacceptable, thus making this model unsuitable despite its high precision.

Decision Tree (without class weights) performed well in terms of both precision and recall, but it did not achieve the same level of balanced accuracy and overall performance

Table 3. Evaluation results of models.

Model		Confusion matrix				Evaluation metrics			
		TP	FP	FN	TN	Balanced accuracy	Precision	Recall	Modified F1 score
Logistic regression	without class weights	94	12	0	0	0.89	0.89	1.00	2.35
	with class weights	67	6	27	6	0.69	0.92	0.71	2.01
Decision Tree	without class weights	82	7	12	5	0.82	0.92	0.87	2.24
	with class weights	80	10	14	2	0.77	0.89	0.85	2.17
Gradient boosting		93	11	1	1	0.89	0.89	0.99	2.35
Random forest		93	10	1	2	0.90	0.90	0.99	2.36
Balanced Random Forest		53	4	41	8	0.58	0.93	0.56	1.75
Bagging Classifier		89	4	10	3	0.87	0.96	0.90	2.32
Balanced bagging Classifier		61	4	33	8	0.65	0.94	0.65	1.92
Easy Ensemble Classifier		65	4	29	8	0.69	0.94	0.69	1.99

as Random Forest. The slight reduction in recall and the presence of more FNs compared to Random Forest led to its rejection. Decision Tree (with class weights) saw a decrease in recall and an increase in FNs, similar to the logistic regression with class weights. The trade-off between precision and recall made it less reliable for predicting all potential crash incidents accurately.

Gradient Boosting exhibited strong performance with high recall and balanced accuracy. However, it did not surpass Random Forest in overall metrics, and the presence of more false positives compared to Random Forest was a drawback. Balanced Random Forest struggled significantly with recall, leading to many false negatives. Its poor balanced accuracy indicated that it was not handling the imbalanced dataset effectively, making it unsuitable for this application.

Bagging Classifier showed a strong performance with high precision and recall, closely matching Random Forest. However, it had slightly more FNs and did not achieve the highest balanced accuracy, leading to its rejection in favor of Random Forest. Balanced Bagging Classifier had high precision but low recall, resulting in numerous FNs. Its moderate balanced accuracy reflected its difficulties in handling imbalanced data, making it less effective for crash prediction. Easy Ensemble Classifier also showed good precision but lower recall, similar to the Balanced Bagging Classifier. The significant number of FNs and only moderate balanced accuracy made it less suitable for the critical task of crash prediction.

In summary, while several models showed strengths in specific areas, **Random Forest** was the only model that consistently balanced high precision and recall with the highest balanced accuracy, making it the best choice for accurately predicting crash incidents.

4.1. Critical determinants of AV crash injuries

An analysis of the importance of various features in an AV crash analysis model makes it evident that they have varying degrees of influence on the model's predictions. Understanding this hierarchy helps focus on the ones that most significantly improve road safety and crash prevention strategies. **Figure 3 depicts the scores of the features that are most important in the random forest model** and reveals that the **damage sustained** by an AV is the most significant predictor. The **company** that manufactured or operates the AV also plays a major role and highlights the differences in technology, safety features, or operational protocols between companies. The **type of vehicle** involved is third in importance, which makes sense given that vehicle types have different safety features and, therefore, behave differently in crashes. **Collision type** is also a noteworthy predictor, as different types of impacts are associated with different crash outcomes. **Movement patterns** of the AV suggest that the dynamics of a vehicle's movements prior to a crash are predictive of the outcome. Less important but still notable is the road type, as it can influence crash likelihood and severity due to speed limits, traffic density, and design.

Intersection geometry and lighting are moderately important features related to the driving environment that play a less-critical role in the driving environment than vehicle-related factors. The presence and type of traffic control signals and parking provisions have a small, yet still relevant, influence on the outcome. Other features such as the

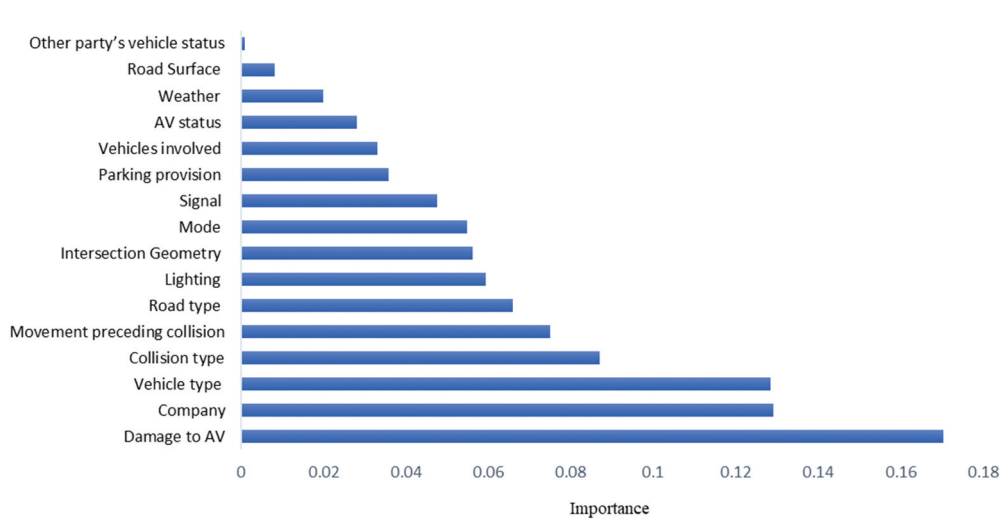


Figure 3. Feature importance score of random forest model.

number of vehicles involved, the status of the AV and the other party's vehicle, and weather and road surface conditions contribute less to the model, indicating that while they do affect crash predictions, they are less impactful than other features.

4.2. Integrated analysis of AV crash data and predictive modeling insights

This section synthesizes empirical data with findings from predictive models to derive a multifaceted understanding of the factors influencing AV crash injuries. It elucidates the complex relationships between vehicle damage, company-specific crash rates, vehicle types, collision types, pre-collision movements, and road types, and the likelihood of injury in AV crashes by correlating the original data set's statistical patterns with model-derived feature-importance scores. Figure 4 presents the statistical critical determinants of AV crash injuries.

4.2.1. Vehicle damage

The distribution of vehicle damage shows a varied pattern in crashes involving AVs that resulted in injuries. No AV damage was reported in 26% of the cases, which could indicate that the injuries were due to indirect causes such as sudden stops; minor damage was shown to be the most common at 44%, suggesting that significant injuries can occur even in low-impact crashes; and moderate damages accounted for 19%, reinforcing the idea that as damage severity increases, so does the likelihood of injury, although not necessarily linearly. Lastly, major vehicle damage was sustained in 11% of the injury cases, highlighting the potential for serious injuries in high severity impacts and underscoring the importance of robust vehicle safety features.

The damage sustained by the AV plays a crucial role in the model, with the pattern of damage observed in crashes presenting a multifaceted picture. The data suggests that even when the damage is minor, there is a significant incidence of injury, which could point to the need for improved safety features that protect

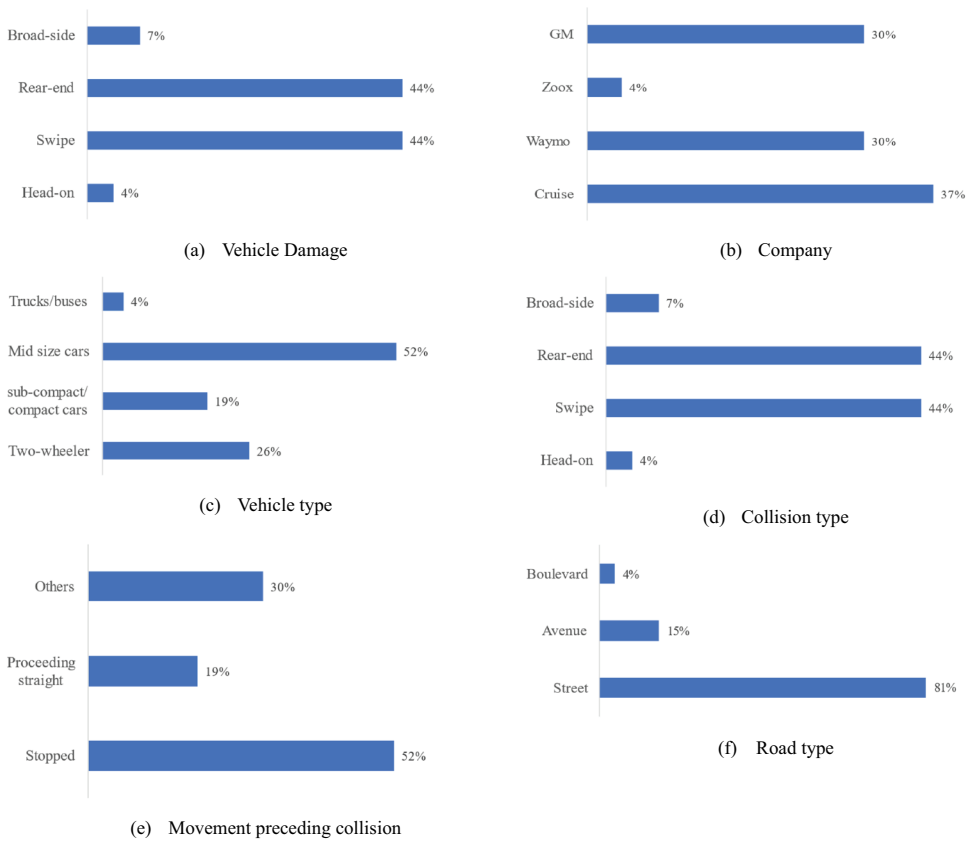


Figure 4. Statistics of critical determinants of AV crash injuries.

occupants despite the extent of the vehicle's damage. The non-linear relationship between damage severity and injury likelihood indicates that even minor vehicle damage can be critical, hence the model's reliance on this feature as a predictor. This result is consistent with the intuitive relationship between the force of impact and the extent of vehicle damage. However, it is important to note that even crashes resulting in minor damage can cause significant injuries, such as whiplash, particularly in rear-end collisions. This non-linear relationship suggests that safety measures need to account for the potential for injury even in low-damage incidents.

4.2.2. Company

The analysis revealed that the AV manufacturer is a critical determinant of crash outcomes. Differences in the technologies and safety protocols used by different manufacturers likely contribute to this finding. The differing rates of injury-related crashes among companies like Cruise, GM, and Waymo suggest that specific factors related to each company's approach – such as safety policies, vehicle design, the types of vehicles used, and technology levels – may influence crash outcomes. While

‘company’ is used as a variable in our model, it should be understood as a proxy for these underlying factors rather than the company itself influencing crash severity. The differences in technology maturity, safety protocols, and operational behaviors are likely to drive the variations in crash outcomes across different manufacturers. The distribution of crashes resulting in injuries among different companies shows that the incidence rates for Cruise, GM, and Waymo are similar, with a 37%, 30%, and 30% share, respectively. Zoox appears to have a notably lower rate at 4%, which could reflect different operational areas, volumes, or possibly more effective safety measures. These figures provide insights into the relative safety performance of each company’s AV technology and could be critical for consumer information and regulatory focus.

4.2.3. Vehicle type

The findings indicate that vehicle type plays a significant role in crash severity. Larger vehicles, such as trucks and SUVs, tend to provide better protection for occupants, which can reduce the likelihood of severe injuries. Smaller vehicles, by contrast, may result in higher injury risks due to their lower mass and structural limitations. This reinforces the importance of vehicle design and crashworthiness in minimizing injury risks. The variations in injury-related crashes across different vehicle types are indicative of the inherent risk associated with each category.

The distribution of vehicle types involved in crashes with injuries reveals a significant variation in the risk profile of different vehicles. Two-wheelers, which are inherently less protective due to their open design, account for 26% of injury-related crashes, suggesting that the drivers/riders are more vulnerable to injuries than those in vehicles with more structural protection. Sub-compact and compact cars follow at 19%, which may reflect their relatively smaller size and possibly less impact-absorbing space compared to larger vehicles. Mid-size cars represent the majority at 52%, possibly due to the higher number of them on the road or a variety of other factors that could contribute to a higher incidence of injuries. Trucks and buses have the smallest share at 4%, which could be due to their larger size and mass providing more protection in a crash; it’s also possible that they are involved in fewer crashes that lead to injuries. These figures provide valuable insights for traffic safety analysis and vehicle design considerations.

4.2.4. Collision type

The distribution of injuries across different collision types is a critical factor in crash severity prediction. Injuries from head-on collisions are relatively rare at 4%, suggesting either effective preventive measures or their lower occurrence, while sideswipe and rear-end collisions are the most common causes of injuries, each counting for 44% of such incidents and indicating a high risk and frequency, likely due to issues like inattentive driving and insufficient spacing between vehicles. Broad-side collisions are also a significant risk but less common at 7%, mainly occurring at intersections.

Rear-end collisions, often associated with sudden braking or stop-and-go traffic, are linked to injuries such as whiplash, even at lower speeds. Sideswipe collisions, typically occurring during lane changes or merging, can result in more severe impacts depending on the speed and angle of the crash. The model’s focus on these collision types highlights the importance of preventative safety features to mitigate their high rates of occurrence and the associated risk of injury. The dynamics of each collision type influence the nature

and **severity of injuries**, underscoring the need for **targeted** safety interventions. In contrast, head-on collisions are rare and may warrant a lower risk profile due to their infrequency in injury cases. However, broad-side collisions, particularly at intersections, pose significant risks due to the severity of these impacts. These findings underline the critical role of safety features and attentive driving in mitigating injury risks across various types of collisions.

4.2.5. *Movement preceding collision*

The breakdown of crashes with injuries by the movement preceding the collision indicates that the **majority of injuries (52%) occur when vehicles are stopped** and illustrates the vulnerability of stationary vehicles to being rear-ended or involved in collisions despite not being in motion. **In contrast, only 19% of injury-related crashes occur when vehicles are travelling in a straight line**, possibly due to the predictable nature of this movement. Finally, other categories that encompass **all other types** of movements represent **30% of the crashes that result in injuries**, suggesting a higher risk associated with less common or more complex driving maneuvers.

The **vulnerability of stopped vehicles** to rear-end collisions reflects limitations in AV response systems during sudden decelerations or when the vehicle is stationary. Rear-end collisions often occur in heavy traffic or congested areas, where AVs may struggle to anticipate the actions of manually driven vehicles behind them. This highlights the need for more advanced AV technologies capable of detecting and predicting the movements of surrounding vehicles, even when the AV is not in motion. The predominance of injuries when vehicles are stopped supports this, as AVs are more susceptible to crashes in these situations, which the model seems to account for. In contrast, the lower incidence of injuries when vehicles are proceeding straight may be due to the predictability and control associated with this movement, which the model factors in as a lower-risk scenario. The predictability of straight-line movement allows AV systems to maintain better control, thus reducing injury outcomes. However, the higher risk associated with more complex driving maneuvers – such as lane changes or turning – underscores the importance of improving AV navigation systems to handle these scenarios safely. This data points to the need for enhancing AV performance in high-risk situations, such as when the vehicle is stationary or engaged in less predictable maneuvers. It also underscores the importance of targeted safety interventions and protective measures for all forms of vehicle movement.

4.2.6. *Road type*

The road type is a significant factor in the analysis of AV crash injuries. Crashes resulting in injuries predominantly occur on streets, likely due to higher traffic, frequent stops, and pedestrian activity. Avenues account for 15%, potentially due to their wider roads that facilitate higher speeds that can increase crash severity. Boulevards, at 4%, have the fewest incidents, possibly because their design allows for safer high-speed travel and fewer stops. The higher incidence of crashes on streets can be attributed to the complex driving environments in urban settings, where frequent stops, pedestrian crossings, and interactions with non-automated vehicles are common. Streets often have a higher density of unpredictable elements, such as pedestrians, cyclists, and traffic signals, all of which require rapid decision-making from AV systems. This observation invites a closer

examination of how AVs are integrated into the urban fabric and suggests that AV navigation systems may require further sophistication to handle the complexity and unpredictability of street-level traffic.

In contrast, avenues and boulevards, with their wider lanes and smoother traffic flow, may present fewer hazards, though they still pose risks due to higher vehicle speeds. These findings indicate that while avenues and boulevards have fewer unpredictable elements, the potential for higher-speed crashes remains a concern. Therefore, refining AV systems to better handle both high-density urban traffic and higher-speed roads is crucial for minimizing injury outcomes. Additionally, integrating AV technology more seamlessly into the urban infrastructure, such as improving traffic signal communication and pedestrian detection systems, could help reduce the number of crashes and injuries occurring on streets. This highlights the need for advanced AV technologies that can better adapt to urban environments where quick adjustments and anticipation of potential collisions are essential.

4.3. Comparison with previous studies

Our findings align with and expand upon several key insights from prior research on AV crashes. Xu et al. (2019) identified rear-end collisions, one-way roads, and roadside parking as significant factors in AV crash severity. Our study corroborates the importance of rear-end collisions but adds that sideswipe collisions are also critical predictors of injury, offering a more nuanced understanding of crash dynamics. Similarly, Das et al. (2020) found that rear-end and sideswipe collisions, along with turning movements and poor lighting, contributed to severe injuries. While our results affirm the importance of rear-end and sideswipe collisions, we place less emphasis on environmental factors like lighting, focusing instead on vehicle-related factors such as AV damage and vehicle type, which we found to be more significant.

Ye et al. (2021) emphasized poor lighting as a key contributor to injury in AV crashes. However, our study found that vehicle damage and collision types, rather than lighting conditions, were the stronger predictors of injury outcomes. This suggests that improvements in AV technology may have reduced the impact of poor lighting on crash severity. Wang et al. (2019) highlighted that crashes are more severe when the AV is at fault and when they occur on highways. While our findings also show that vehicle type and crash responsibility are important, we further demonstrate that vehicle damage is a critical predictor, which adds an important dimension to understanding injury outcomes in AV crashes.

Leilabadi and Schmidt (2019) pointed out that wet roadway surfaces increase the likelihood of AV crashes. In contrast, our results indicate that road conditions, while relevant, are less significant in predicting injury outcomes compared to vehicle-related factors. This difference may reflect technological improvements in AVs' ability to handle adverse weather conditions. Zhang and Xu (2021) identified nighttime driving, multiple-vehicle accidents, and high-density traffic as major contributors to crash severity. Our findings support the importance of multiple-vehicle accidents but suggest that factors such as AV manufacturer and vehicle type are more critical to injury outcomes than driving conditions like nighttime or traffic density. Lastly, Ding et al. (2023) emphasized weather conditions and traffic incidents. While our study recognizes the role of collision

types, we found that AV manufacturer and vehicle type had a greater influence on injury outcomes, suggesting that design and technology factors now play a larger role than external conditions in determining crash severity.

5. Conclusion

AVs present an unprecedented opportunity to mitigate the leading cause of mortality worldwide by **eliminating the human error component in driving and enhancing road safety**. The main goal of this research was to create a predictive model using machine learning to distinguish between injury and non-injury AV accidents. It also aimed to prioritize the factors influencing crash outcomes to gain insights into their impact on these incidents. This study addresses a critical research gap by utilizing the most comprehensive and up-to-date dataset on AV crashes, spanning from 2014 to 2024. In a field where technology is rapidly advancing, it is essential to use recent data to ensure that predictive models and safety analyses are relevant to current AV systems. AV crash data from the CA DMV was utilized to examine 530 incidents and 16 variables, transcending mere quantitative examination and culminating in a nuanced understanding of crash dynamics.

The crash prediction models developed in this research despite significant challenges related to imbalanced data, offer significant insights into the factors contributing to AV incidents. The problem of the imbalanced data, which is typical in real-world scenarios where the number of non-injury crashes significantly outweighs injury crashes, was solved by calculating class weights, to assign higher importance to the minority class to prevent model bias. The comprehensive analysis of machine learning models for crash prediction emphasizes the complexity of selecting an appropriate model for safety and injury prevention in the context of AVs. **The random forest emerged as the model of choice** due to its **superior balance between sensitivity and specificity**, coupled with the lowest number of FNs – a critical factor in scenarios where avoiding missed injury predictions is paramount. The detailed assessment of feature importance and the integration of these insights with empirical AV crash data identified several critical determinants of AV crash injuries. **Vehicle damage, manufacturing company, vehicle type, collision type, pre-collision movement, and road type were recognized as influential factors in predicting crash severity**. Among these, vehicle damage stands out as a primary predictor, suggesting the significance of vehicle integrity and the non-linear relationship between damage severity and injury likelihood. The predominance of injuries on streets, the vulnerability of stopped vehicles, and the high risk associated with certain movements and collision types all feed into this hierarchy, providing a clear direction for prioritizing improvements. The observation that most injuries occur on streets invites a closer examination of how **AVs are integrated into the urban fabric**. Collectively, these insights underscore the need for a holistic approach to road safety that is adaptive to the varied conditions and challenges posed by the evolving landscape of autonomous mobility.

The study's limitations are rooted in its reliance on geographically limited and potentially incomplete data from California DMV reports, which may not accurately

reflect global AV performance due to different driving conditions and reporting standards. Furthermore, biases resulting from inconsistent detail levels among submissions can further skew the data and potentially impact the accuracy of predictions. Future research in the realm of AV crash analysis could benefit from expanding the geographical scope of data collection, allowing for the examination of AV performance across diverse traffic environments and regulations. Incorporating richer data sets, potentially augmented with more detailed telematics and sensor information, could yield a deeper understanding of the nuanced dynamics at play during AV incidents.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

The data that support the findings of this study are openly available in the CA DMV database. Specifically, the data were extracted from the Autonomous Vehicle Collision Reports section, which provides detailed and publicly accessible information on collisions involving AVs. The dataset can be accessed by visiting the following link: <https://www.dmv.ca.gov/portal/vehicle-industry-services/autonomous-vehicles/autonomous-vehicle-collision-reports/>

References

- Almaskati, D., Kermanshachi, S., & Pamidimukkala, A. (2024). A review of factors affecting severity of autonomous vehicle crashes. *International Conference on Transportation and Development 2024* (pp. 755–766), Atlanta, Georgia.
- Almaskati, D., Pamidimukkala, A., Kermanshachi, S., Rosenberger, J., & Foss, A. (2024). Investigation of the impacts of the deployment of autonomous vehicles on first responders. *Smart and Resilient Transportation*. <https://doi.org/10.1108/SRT-05-2024-0005>
- Ashraf, M. T., Dey, K., Mishra, S., & Rahman, M. T. (2021). Extracting rules from autonomous-vehicle-involved crashes by applying decision tree and association rule methods. *Transportation Research Record*, 2675(11), 522–533. <https://doi.org/10.1177/03611981211018461>
- Channamallu, S. S., Almaskati, D., Kermanshachi, S., & Pamidimukkala, A. (2024a). Autonomous vehicle safety: A comprehensive analysis of crash injury determinants. *International Conference on Transportation and Development 2024* (pp. 767–779), Atlanta, Georgia.
- Channamallu, S. S., Almaskati, D., Kermanshachi, S., & Pamidimukkala, A. (2024b). Enhancing AV safety: A bagging classifier approach for predicting crash outcomes. *International Conference on Transportation and Development 2024* (pp. 538–549), Atlanta, Georgia.
- Channamallu, S. S., Kermanshachi, S., & Pamidimukkala, A. (2023). Impact of autonomous vehicles on traffic crashes in comparison with conventional vehicles. *International Conference on Transportation and Development 2023*, Austin, Texas (pp. 39–50).
- Das, S., Dutta, A., & Tsapakis, I. (2020). Automated vehicle collisions in California: Applying Bayesian latent class model. *IATSS Research*, 44(4), 300–308. <https://doi.org/10.1016/j.iatssr.2020.03.001>

- Ding, S., Abdel-Aty, M., Wang, D., Wang, Z., Zheng, O., Barbour, N., & Yuan, R. (2023). Exploratory analysis of the crash severity between vehicular automation (SAE L2-5) with multi-source data. <https://arxiv.org/abs/2303.17788>
- Durić, P., & Miladinov-Mikov, M. (2008). Some characteristics of drivers having caused traffic accidents. *Medical Review*, 61(9–10), 464–469. <https://doi.org/10.2298/MPNS0810464D>
- Favarò, F. M., Nader, N., Eurich, S. O., Tripp, M., Varadaraju, N., & Hu, X. (2017). Examining accident reports involving autonomous vehicles in California. *PLOS ONE*, 12(9), e0184952. <https://doi.org/10.1371/journal.pone.0184952>
- Guo, X., & Zhang, Y. (2022). Maturity in automated driving on public roads: A review of the six-year autonomous vehicle tester program. *Transportation Research Record*, 2676(11), 352–362. <https://doi.org/10.1177/03611981221092720>
- Houseal, L. A., Gawesh, S. M., Dadvar, S., & Ahmed, M. M. (2022). Causes and effects of autonomous vehicle field test crashes and disengagements using exploratory factor analysis, binary logistic regression, and decision trees. *Transportation Research Record*, 2676(8), 571–586. <https://doi.org/10.1177/03611981221084677>
- Khattak, Z. H., Fontaine, M. D., & Smith, B. L. (2020). Exploratory investigation of disengagements and crashes in autonomous vehicles under mixed traffic: An endogenous switching regime framework. *IEEE Transactions on Intelligent Transportation Systems*, 22(12), 7485–7495. <https://doi.org/10.1109/TITS.2020.3003527>
- Kutela, B., Avelar, R. E., & Bansal, P. (2022). Modeling automated vehicle crashes with a focus on vehicle at-fault, collision type, and injury outcome. *Journal of Transportation Engineering, Part A: Systems*, 148(6), 04022024. <https://doi.org/10.1061/JTEPBS.0000680>
- Kutela, B., Das, S., & Dadashova, B. (2022). Mining patterns of autonomous vehicle crashes involving vulnerable road users to understand the associated factors. *Accident Analysis & Prevention*, 165, 106473. <https://doi.org/10.1016/j.aap.2021.106473>
- Leilabadi, S. H., & Schmidt, S. (2019). In-depth analysis of autonomous vehicle collisions in California. 2019 IEEE Intelligent Transportation Systems Conference (ITSC), Auckland, New Zealand (pp. 889–893). IEEE.
- Li, Z., Chen, W., & Pei, D. (2018). Robust and unsupervised KPI anomaly detection based on conditional variational autoencoder. Proceedings of the 2018 IEEE 37th International Performance Computing and Communications Conference (IPCCC), Orlando, FL, USA (pp. 1–9). 17–19 November 2018
- Liu, Q., Wang, X., Wu, X., Glaser, Y., & He, L. (2021). Crash comparison of autonomous and conventional vehicles using pre-crash scenario typology. *Accident Analysis & Prevention*, 159, 106281. <https://doi.org/10.1016/j.aap.2021.106281>
- Pamidimukkala, A., Kermanshachi, S., & Patel, R. (2023). An exploratory analysis of crashes involving autonomous vehicles. International Conference on Transportation and Development 2023, Austin, Texas (pp. 343–350).
- Patel, R. K., Etminani-Ghasrodashti, R., Kermanshachi, S., Rosenberger, J. M., & Foss, A. (2023). Exploring willingness to use shared autonomous vehicles. *International Journal of Transportation Science and Technology*, 12(3), 765–778. <https://doi.org/10.1016/j.ijtst.2022.06.008>
- Petrović, Đ., Mijailović, R., & Pešić, D. (2020). Traffic accidents with autonomous vehicles: Type of collisions, manoeuvres and errors of conventional vehicles' drivers. *Transportation Research Procedia*, 45, 161–168. <https://doi.org/10.1016/j.trpro.2020.03.003>
- Ren, W., Yu, B., Chen, Y., & Gao, K. (2022). Divergent effects of factors on crash severity under autonomous and conventional driving modes using a hierarchical Bayesian approach. *International Journal of Environmental Research and Public Health*, 19(18), 11358. <https://doi.org/10.3390/ijerph191811358>
- Schanz, A. E., & Hamann, R. (2012). Towards “vision zero”. Proceedings of the SAE 2012 World Congress & Exhibition SAE International, Detroit, MI, USA.
- Sing, S. (2018). *Critical reasons for crashes investigated in the national motor vehicle crash causation survey*. National Center for Statistics and Analysis.

- Sinha, A., Vu, V., Chand, S., Wijayaratna, K., & Dixit, V. (2021). A crash injury model involving autonomous vehicle: Investigating of crash and disengagement reports. *Sustainability*, 13(14), 7938. <https://doi.org/10.3390/su13147938>
- Song, Y., Chitturi, M. V., & Noyce, D. A. (2021). Automated vehicle crash sequences: Patterns and potential uses in safety testing. *Accident Analysis & Prevention*, 153, 106017. <https://doi.org/10.1016/j.aap.2021.106017>
- Wang, S., Li, Z., & Guo, Y. (2019). Exploring the mechanism of crashes with automated vehicles using statistical modeling approaches. *PLOS ONE*, 14(3), e0214550. <https://doi.org/10.1371/journal.pone.0214550>
- Xu, C., Ding, Z., Wang, C., & Li, Z. (2019). Statistical analysis of the patterns and characteristics of connected and autonomous vehicle involved crashes. *Journal of Safety Research*, 71, 41–47. <https://doi.org/10.1016/j.jsr.2019.09.001>
- Ye, W., Wang, C., Chen, F., Yan, S., & Li, L. (2021). Approaching autonomous driving with cautious optimism: Analysis of road traffic injuries involving autonomous vehicles based on field test data. *Injury Prevention*, 27(1), 42–47. <https://doi.org/10.1136/injuryprev-2019-043402>
- Yuan, Q., Xu, X., Wang, T., & Chen, Y. (2022). Investigating safety and liability of autonomous vehicles: Bayesian random parameter ordered probit model analysis. *Journal of Intelligent and Connected Vehicles*, 5(3), 199–205. <https://doi.org/10.1108/JICV-04-2022-0012>
- Zhang, J., & Xu, C. (2021). Investigating the typical scenarios and contributory factors to crash severity of autonomous vehicle involved collisions using association rule analysis. Proceedings of the 100th Annual Meeting of the Transportation Research Board (pp. 5–29), Washington, DC, USA.
- Zhu, S., & Meng, Q. (2022). What can we learn from autonomous vehicle collision data on crash severity? A cost-sensitive CART approach. *Accident Analysis & Prevention*, 174, 106769. <https://doi.org/10.1016/j.aap.2022.106769>